

Lesson 3: Resistance and Ohm's law

Electric Circuits · Year 10 · 50 minutes · AQA 6.2.1.3-4

Why this lesson is here

Two devices on the same supply draw different currents. Their resistances differ: $R = V/I$ compares p.d. with current.

Learning objective

Calculate resistance and explain why a fixed resistor can obey $V = IR$.

Core teaching point

For a fixed resistor at constant temperature, current is directly proportional to p.d. Resistance is measured in ohms (ohm). Heating can change resistance.

Equipment and preparation

PhET lab; virtual-lab worksheet pp. 12-16; low-voltage supply, fixed resistors and meters for demonstration.

Lesson sequence

Time	Phase	Teacher move and evidence
0-5	Retrieval	Convert milliampere to ampere.
5-16	Model	Derive $R = V/I$ and work through a unit-aware calculation.
16-32	Virtual Lab 3	Measure and compare R with one change at a time.
32-43	Practice	Tasks 1, 2 and 4; distinguish line shape and gradient.
43-50	Check	Four-mark question; discuss constant temperature.

Workbook task guidance

Task 1: Calculate R for (a) 12 V, 0.60 A; (b) 3.0 V, 0.15 A. State the unit.

Expected: (a) 20 ohm; (b) 20 ohm.

Task 2: Explain why the phrase 'a resistor obeys Ohm's law' needs the words 'at constant temperature'.

Expected: A fixed resistor has constant R only if its temperature is held constant.

Task 3: Virtual Lab 3: complete the resistance investigation on existing worksheet pp. 12-16. Change one variable at a time and compare calculated R values.

Expected: Calculated R should remain near constant for the same fixed resistor at the same temperature; explain discrepancies with measurement and heating.

Task 4: Sketch current (y) against p.d. (x) for a fixed resistor and state what the gradient tells you qualitatively.

Expected: Straight line through the origin; with I on y and V on x, steeper means smaller R.

Worked example

A resistor has 6.0 V across it and 0.30 A through it. $R = V/I = 6.0/0.30 = 20$ ohm.

AQA-style checkpoint

[4 marks] A fixed resistor has 2.0 V across it and carries 0.10 A. Calculate its resistance and predict the current at 6.0 V if its temperature stays constant.

Mark guidance: $R = V/I$ (1) = $2.0/0.10 = 20$ ohm (1). $I = V/R$ (1) = $6.0/20 = 0.30$ A (1).

Anticipated misconception

A larger current does not always mean lower resistance: compare at the same p.d. and temperature.

Assessment and next step

Collect one sentence of reasoning and one complete calculation or graph label. Start the next lesson by reteaching the most common error; students who are secure should tackle a changed numerical context.